Chapter 1 — Java and the JVM

What happens between typing `javac` (the Java compiler) and seeing output: setting up a JDK (Java Development Kit — compiler plus developer tools), three ways to **compile** (translate source code into a form a machine can execute) and run programs, what is inside a `.class` file, the Just-In-Time compiler and the garbage collector at work, `jshell`, and how Java versions and long-term-support releases work.

All code lives in `code/`, compiled with `javac --release 25` (the book targets **Java 25 LTS**) and run on JDK 26; machine-dependent output (timings, memory sizes) is flagged. Exam-level traps (class structure, `main` signatures, packages, GC eligibility): the OCP repo's Chapter 1 — Building Blocks.

1.1 What Java is

What it is

Java is a **general-purpose, statically typed, object-oriented programming language** plus a **runtime platform** (the Java Virtual Machine) that runs compiled Java programs. A **runtime** is the software that must be present while a program runs, executing it and providing services such as memory management.

- *General-purpose*: not tied to one domain — banking back-ends, Android apps (historically), big-data engines, command-line tools.
- *Statically typed*: every variable has a type known at compile time, so the compiler rejects `int n = "five";` before the program runs.
- *Object-oriented*: code is organised into **classes** (blueprints) and **objects** (instances built from them). Modern Java also supports functional style — lambdas (short unnamed functions passed as values) and streams (pipelines that filter and transform data), Chapters 11–12.

Compiled Java runs on any machine with a JVM (Windows, macOS, Linux, containers — isolated packages such as Docker images that bundle an app with everything it needs): *"write once, run anywhere"*.

A short history

Year	Event
1991–96	James Gosling's team at Sun starts the "Green Project" (language: **Oak**); renamed **Java** and announced May 1995 (browser "applets" make it famous); JDK 1.0 January 1996.
2004	J2SE 5.0: generics, enhanced `for`, autoboxing, enums, annotations, varargs — the biggest language update until Java 8.
2006–10	Sun open-sources the JDK as **OpenJDK** (GPL v2 + Classpath Exception); Oracle acquires Sun (2010).
2014	Java 8: lambdas, streams, `java.time`. Still widely deployed.
2017	Java 9: the module system (JPMS — groups packages into named, sealed-off modules) and `jshell`.
2018–21	Java 10 starts the **six-month cadence** (March/September); Java 17 LTS (2021), LTS now every **two years**.
2023	Java 21 LTS: virtual threads, pattern matching for `switch`, record patterns.
2025	**Java 25 LTS** (16 Sept 2025): compact source files and instance `main`, flexible constructor bodies, module imports, scoped values.
2026	Java 26 (March) and Java 27 (15 Sept 2026) — non-LTS.

Design traits

Trait	In practice	Caveat
Platform independence	Bytecode (the JVM's portable instruction format) runs on any OS/CPU with a JVM.	Target still needs a JVM or a native-image build (pre-compiled executable).
Static typing	Compiler catches type errors; IDEs refactor safely.	More declarations than Python/JS (softened by `var`, records).
Automatic memory management	A garbage collector frees unused objects; no `free()` / `delete`.	GC costs CPU/memory; you can still leak by keeping references (handles to objects).
Backward compatibility	Java 8 bytecode usually runs on Java 25.	Old APIs linger (`Date`, `Vector`).
Rich standard library	Collections, concurrency, networking, HTTP client, `java.time`, crypto.	Large; finding the modern API takes time.
Performance	The JIT turns hot (frequently run) code into optimised machine code.	Slower start-up/warm-up (mitigated by AOT — ahead-of-time — caches, native images).
Safety	No pointer arithmetic, bounds-checked arrays, verified bytecode.	Small cost.

Where Java is used

Enterprise back-ends (banks, airlines, retailers on Spring Boot / Jakarta EE — **frameworks**, large reusable libraries that structure an app); **big data** (Kafka, Cassandra, Elasticsearch, Hadoop, much of Flink); **developer tools** (IntelliJ IDEA, Eclipse, Maven, Gradle, Jenkins); **Android** (historically the Java *language* and much of its library on its own runtime, ART, not the JVM; Kotlin-first since 2019); Minecraft: Java Edition; Java Card on SIM and bank cards. **Kotlin, Scala, Groovy, Clojure** also compile to JVM bytecode and interoperate with Java libraries.

> **TL;DR**
>
> Java = statically typed, object-oriented language + portable runtime (the JVM). Mature (1996), backward compatible, fast after warm-up, dominant in server-side and data infrastructure.

1.2 The Java ecosystem: SE, EE, OpenJDK and vendors

What it is

Name	What it is
Java SE (Standard Edition)	The *specification* of the language, JVM and core library (`java.lang`, `java.util`, …). Each version is defined through the Java Community Process (the formal process for agreeing changes).
JDK (Java Development Kit)	A concrete *implementation* of Java SE plus tools (`javac`, `jshell`, `jar`, `javadoc`, `jlink`…).
OpenJDK	The open-source project (openjdk.org) where the reference JDK is developed. Almost every downloadable JDK is built from it.
Vendor builds	Oracle JDK, Eclipse Temurin (Adoptium), Amazon Corretto, Azul Zulu, Microsoft Build of OpenJDK, Red Hat, BellSoft Liberica, SAP Machine. Same code base; different support, licensing, extras.
Jakarta EE (formerly Java EE)	Enterprise API specs on top of Java SE (servlets — handle web requests; JPA — persist objects to databases; CDI — wire objects together). Moved from Oracle to the Eclipse Foundation (2017) and renamed.
Java ME / Java Card	Small-device and smart-card editions.
GraalVM	Oracle JDK distribution with the Graal compiler and **Native Image**, which compiles Java ahead of time into a standalone native executable (no JVM needed).

Which vendor should I install?

For learning, any OpenJDK 25 build; **Eclipse Temurin** is a neutral choice. Companies choose by support contract, cloud (Corretto on AWS, Microsoft's build on Azure) or licensing. Oracle JDK 17+ uses the *Oracle No-Fee Terms and Conditions*: free **production** use for a limited period (JDK 25 updates free until September 2028), then later updates need a paid licence. Other vendors' OpenJDK builds are GPL.

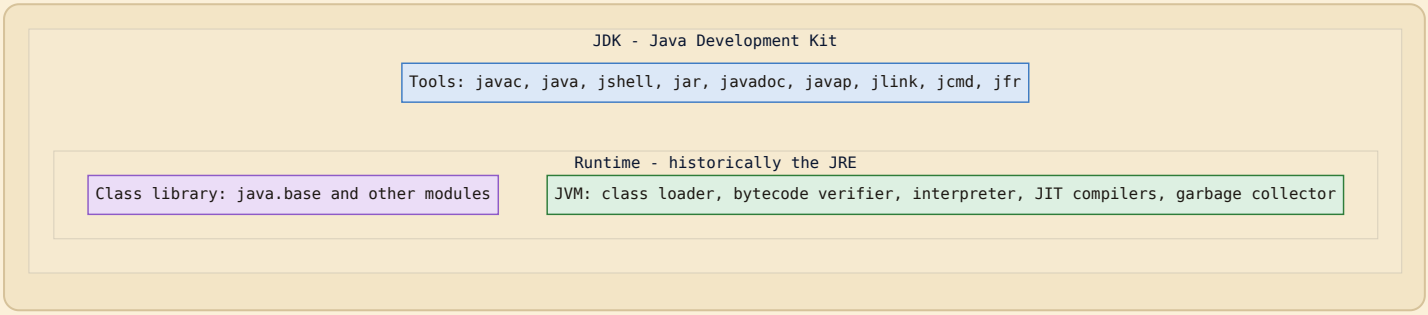
Common mistakes

- Thinking OpenJDK is lower quality than Oracle JDK — same source.
- Confusing Java with **JavaScript**; they share only a name and C-like syntax (JavaScript Handbook).

1.3 JDK, JRE and JVM

What it is

Three nested layers:



Layer	Contains	Who needs it
JVM (Java Virtual Machine)	Abstract machine that loads, verifies and executes bytecode (interpreting and JIT-compiling) and manages memory. **HotSpot** is the JVM in OpenJDK builds.	Every Java program at run time
JRE (Java Runtime Environment)	JVM + standard class library — enough to *run*, not compile.	End users (historically)
JDK	JRE + development tools (`javac`, `jshell`, `jar`, `javadoc`, debuggers, profilers).	Developers — **install this**

Virtual machine here means a program that acts as a CPU with its own instruction set (bytecode) — not a VirtualBox-style whole-computer emulator.

Why the JRE "disappeared"

Since JDK 11, Oracle ships no separate JRE. Java 9's module system introduced `jlink`, which builds a **custom runtime** containing only the modules (named groups of packages, such as `java.base`) your app needs. Some vendors (e.g. Temurin) still publish JRE-only builds, mainly for containers.

A runtime with only `java.base` is enough for `HelloWorld`:

```
jlink --add-modules java.base --strip-debug --no-header-files --no-man-pages --output rt
du -sh rt                       # 44M   (the full JDK folder here is 377M)
rt/bin/java -cp out ch01.hello.HelloWorld
```

Output:

```
Hello, World!
Arguments received: 0
```

In the interview

Q: Difference between JDK, JRE and JVM? The JVM executes bytecode and manages memory; the JRE is JVM + standard library (enough to run); the JDK is JRE + tools like `javac`. Developers install a JDK; production images often use a `jlink`-trimmed runtime.

- *Is the JVM platform-independent?* No — the **bytecode** is portable; each OS/CPU needs its own JVM build.
- *What is HotSpot?* OpenJDK's JVM, named for finding "hot" code and compiling it.
- *Is a JRE enough to compile?* No — you need the JDK (`javac`).

1.4 Installing a JDK and checking it works

How it works

1. **Download** an OpenJDK 25 build (e.g. Temurin from adoptium.net), or use **SDKMAN!** (`sdk list java`, `sdk install java <identifier>`) or an OS package manager (`apt`, Homebrew).
2. **Set** `JAVA_HOME` to the JDK folder and put `$JAVA_HOME/bin` on `PATH` (build tools and IDEs read `JAVA_HOME`). An **environment variable** is a named setting the OS passes to every program; `PATH` lists folders searched for commands. macOS: `export JAVA_HOME=$(/usr/libexec/java_home -v 25)`; Linux: `export JAVA_HOME=/usr/lib/jvm/<jdk-folder>`; Windows: *System Properties → Environment Variables*, add `%JAVA_HOME%\bin` to `Path`.
3. **Verify** in a new terminal:

```
java -version
javac -version
```

Output (on the machine used for this book):

```
openjdk version "26" 2026-03-17
OpenJDK Runtime Environment (build 26+35-2893)
OpenJDK 64-Bit Server VM (build 26+35-2893, mixed mode, sharing)
javac 26
```

"mixed mode" = the JVM both interprets and JIT-compiles; *"sharing"* = it uses Class Data Sharing (a pre-parsed archive of core classes that speeds start-up).

Compiling for an older version: `--release`

`javac --release 25 …` checks code against the Java 25 language **and** API, so a newer JDK cannot let a Java 26 method slip in. That is how this book targets 25 on JDK 26. The older `-source`/`-target` pair does not check the API — prefer `--release`.

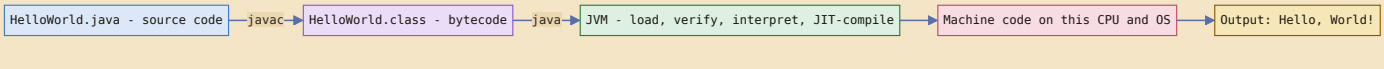
Common mistakes

- Several JDKs installed, with `java` and `javac` on `PATH` from different ones — check both versions.
- `JAVA_HOME` pointing at `bin` instead of the JDK root.
- Installing only a JRE, then `javac`: "command not found".

1.5 Write → compile → run

What it is

Java is **compiled in two stages**: `javac` turns source (`.java`) into platform-neutral **bytecode** (`.class` files); the `java` launcher starts a JVM that loads and runs it.



The program

`code/HelloWorld.java`:

```java
package ch01.hello;

/** The classic first program: one public class, one entry point. */
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Arguments received: " + args.length);
    }
}
```

Part	Meaning
`package ch01.hello;`	The **package** (namespace); must be the first statement. Fully qualified name: `ch01.hello.HelloWorld`. See Chapter 5.
`/** … */`	**Javadoc comment**, turned into HTML API docs by `javadoc`. `//` is a line comment, `/* … */` a block comment.
`public class HelloWorld`	At most one **public** top-level class per file, and the file must be named after it.
`public static void main(String[] args)`	The **entry point**. `public` = callable from anywhere; `static` = no object needed; `void` = returns nothing; `args` = command-line arguments (Chapter 4).
`System.out.println(...)`	Prints a line to **standard output** (the terminal by default). `System` is in `java.lang` (imported automatically — an *import* lets you use a class by its short name); `out` is its static `PrintStream` field (a class-level variable holding a text-writing object).
`;` and `{ }`	Statements end with a semicolon; braces group blocks. Java ignores indentation.

Since Java 25 a class may instead declare `void main()` (no `static`, no parameters) — see 1.10. `public static void main(String[] args)` remains the standard and works in every version.

How to compile and run

```
javac -d out HelloWorld.java            # compile; -d puts classes in out/ch01/hello/
java -cp out ch01.hello.HelloWorld       # run: class path + fully qualified class name
java -cp out ch01.hello.HelloWorld a b c
```

Output:

```
Hello, World!
Arguments received: 0
Hello, World!
Arguments received: 3
```

1. `javac` parses the file, checks types and writes `out/ch01/hello/HelloWorld.class` (folders mirror the package).
2. `java` starts a JVM; `-cp` (class path) says where to look for classes.
3. The JVM loads `ch01.hello.HelloWorld`, finds `main`, and calls it with the arguments.

Reading errors: compile-time vs run-time

`javac` reports compile errors with file, line and a caret:

```
Hello.java:3: error: ';' expected
        System.out.println("hi")
                                ^
```

```
Hello.java:1: error: class Greeting is public, should be declared in a file named Greeting.java
```

```
Hello.java:3: error: incompatible types: String cannot be converted to int
        int n = "five";
                ^
```

Launcher errors happen when `java` cannot find the class:

Command	Error	Fix
`java hello`	`NoClassDefFoundError: hello (wrong name: Hello)` on macOS/Windows (case-insensitive file systems); `ClassNotFoundException: hello` on Linux	Class names are case-sensitive.
`java Hello.class`	`ClassNotFoundException: Hello.class`	Pass the class **name**, not the file name.
`java HelloWorld` (class is in a package)	`Could not find or load main class HelloWorld`	Use the fully qualified name and the right `-cp`.

Packaging a JAR

A **JAR** (Java ARchive) is a ZIP of classes plus a manifest (metadata file); a `Main-Class` entry makes it runnable:

```
jar --create --file hello.jar --main-class ch01.hello.HelloWorld -C out .
java -jar hello.jar one two
```

Output:

```
Hello, World!
Arguments received: 2
```

Real projects let Maven or Gradle build JARs (Chapter 18).

Common mistakes

- File name ≠ public class name (case matters: `helloworld.java` fails for `HelloWorld`).
- Missing/wrong `-cp`, or running from inside the package folder.
- Misspelt `main` signature (e.g. `String args` without `[]`) in a traditional class — compiles, but the launcher reports no suitable main method.

TL;DR

`javac` → bytecode; `java -cp <dir> <package.Class>` → JVM runs it. `-d out` puts `ch01.hello.HelloWorld` at `out/ch01/hello/HelloWorld.class`. One public top-level class per file, named like the file.

1.6 Bytecode and class files

What it is

Bytecode is the JVM's instruction set: compact one-byte opcodes (operation codes such as `iload`, `iadd`, `invokevirtual`) for a **stack machine** — instructions push values onto an operand stack and pop them to compute. A `.class` file holds one class's bytecode plus a **constant pool** (names, strings, method references) and metadata.

Why it matters

- It is the unit of portability: the same `.class` runs on every JVM.
- The JVM verifies it before running (no forged pointers, no operand-stack overflow/underflow, no jumps into the middle of an instruction).
- Tools — profilers (measure time/memory), mocking libraries (fake objects for tests), Spring, Hibernate — read or generate bytecode at run time.

Looking inside with `javap`

`code/Calculator.java`:

```java
package ch01.bytecode;

public class Calculator {
    static int add(int a, int b) {
        return a + b;
    }

    public static void main(String[] args) {
        int total = add(2, 3);
        System.out.println("2 + 3 = " + total);
    }
}
```

`javap -c -cp out ch01.bytecode.Calculator` (the JDK's disassembler):

```
public class ch01.bytecode.Calculator {
  public ch01.bytecode.Calculator();
    Code:
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return

  static int add(int, int);
    Code:
         0: iload_0
         1: iload_1
         2: iadd
         3: ireturn

  public static void main(java.lang.String[]);
    Code:
         0: iconst_2
         1: iconst_3
         2: invokestatic  #7                  // Method add:(II)I
         5: istore_1
         6: getstatic     #13                 // Field java/lang/System.out:Ljava/io/PrintStream;
         9: iload_1
        10: invokedynamic #19,  0             // InvokeDynamic #0:makeConcatWithConstants:(I)Ljava/lang/String;
        15: invokevirtual #23                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        18: return
}
```

1. `javac` added a **default constructor** `Calculator()` (a constructor sets up a new object) that calls `Object`'s constructor — you did not write it.
2. `add`: `iload_0`/`iload_1` push the two `int` parameters, `iadd` pops both and pushes the sum, `ireturn` returns it (`i` = int).
3. `main`: pushes 2 and 3, calls `add` (`(II)I` = two ints in, int out), stores the result in local slot 1.
4. String concatenation compiles to `invokedynamic … makeConcatWithConstants` (since Java 9), letting the JVM choose the fastest strategy at run time.

Class-file version

Every class file starts with the magic number `CAFEBABE` (fixed bytes identifying the file type), then a **major version**: major = 44 + Java version.

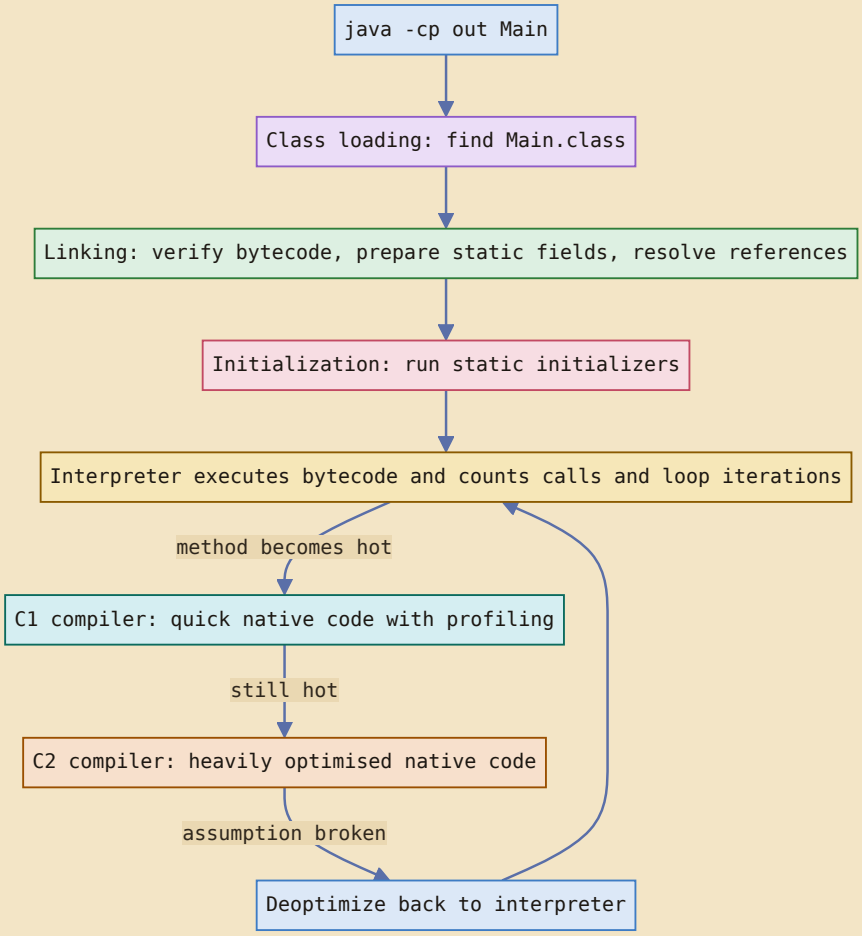
Java	8	11	17	21	25	26
Major version	52	55	61	65	**69**	70

```
javap -v -cp out ch01.bytecode.Calculator | grep major   # major version: 69
xxd out/ch01/bytecode/Calculator.class | head -1          # cafe babe 0000 0045 ...  (0x45 = 69)
```

An older JVM refuses newer class files with `UnsupportedClassVersionError` ("compiled by a more recent version of the Java Runtime (class file version 69.0), this version … only recognizes class file versions up to 65.0" — i.e. compiled by Java 25, run on 21). Fix: run on a newer JVM or compile with `--release <older>`.

1.7 How the JVM runs your code: class loading, interpreter, JIT

How it works



1. **Class loading.** *Class loaders* turn `.class` bytes into classes, lazily (the first time a class is actively used). Three built-in loaders: **bootstrap** (core `java.base`), **platform** (other JDK modules), **application/system** (your class path). Each asks its parent first (*parent delegation*), so you cannot replace `java.lang.String`.
2. **Linking.** *Verification* checks the bytecode is type-safe and well-formed; *preparation* allocates static fields with default values; *resolution* turns symbolic names (e.g. `java/io/PrintStream.println`) into direct references (lazily).
3. **Initialization.** Static field initializers and `static { }` blocks run, once per class.
4. **Execution.** The **interpreter** runs bytecode instruction by instruction — instant start, slow — counting how often each method and loop runs.
5. **JIT compilation.** *Just-In-Time* compilation converts **hot** methods to native code while the program runs. HotSpot uses **tiered compilation** (default since Java 8):

Tier	Who	What
0	Interpreter	Runs bytecode, gathers counts
1–3	**C1** ("client") compiler	Compiles fast, light optimisation; tier 3 adds profiling (which types and branches actually occur)
4	**C2** ("server") compiler	Uses the profile for aggressive optimisation

C2's optimisations: **inlining** (copy a small method's body into the caller), **loop unrolling** (repeat the body several times per iteration to cut loop overhead), **escape analysis** (objects that never leave a

method need not live on the heap), **dead-code removal** (delete work whose result is unused).

6. **Speculation and deoptimization.** C2 optimises for what it has *seen* (e.g. "this call always hits `ArrayList`"). If that breaks, the JVM discards the compiled code ("uncommon trap"), returns to the interpreter, and may recompile. This is why a JIT can beat a static compiler: it optimises for the types and branches actually seen.

Watching the JIT

`code/HotLoop.java` calls a tiny method five million times:

```
static long square(long x) { return x * x; }

public static void main(String[] args) {
    long sum = 0;
    for (long i = 0; i < 5_000_000; i++) {
        sum += square(i % 1000);
    }
    System.out.println("sum = " + sum);
}
```

`java -XX:+PrintCompilation -cp out ch01.jit.HotLoop | grep -E "HotLoop|sum ="` (real output; timestamps vary):

```
15    9       3       ch01.jit.HotLoop::square (4 bytes)
15   10       4       ch01.jit.HotLoop::square (4 bytes)
15    9       3       ch01.jit.HotLoop::square (4 bytes)   made not entrant: not used
16   11 %     3       ch01.jit.HotLoop::main @ 4 (43 bytes)
16   12       3       ch01.jit.HotLoop::main (43 bytes)
16   13 %     4       ch01.jit.HotLoop::main @ 4 (43 bytes)
16   11 %     3       ch01.jit.HotLoop::main @ 4 (43 bytes)   made not entrant: OSR invalidation of lower level
19   13 %     4       ch01.jit.HotLoop::main @ 4 (43 bytes)   made not entrant: uncommon trap
sum = 1664167500000
```

Columns: ms since start, compile ID, flags, **tier**, method. `square` was compiled by C1 (tier 3) then C2 (tier 4) in the same millisecond, and the C1 version discarded ("made not entrant"). `%` = **on-stack replacement (OSR)**: `main` runs once but its loop is hot, so the JVM compiles the loop and switches to it *while the method is running*. The last line is a deoptimization.

Start-up and warm-up: AOT caching

Starting in the interpreter makes a Java service slower for its first seconds (**warm-up**). Recent JDKs record an **ahead-of-time (AOT) cache** from a training run:

- JDK 24 (JEP 483 — a *JEP* is a numbered JDK Enhancement Proposal, 1.12): classes loaded and linked ahead of time into a cache.
- JDK 25 (JEP 514): one-step creation — `java -XX:AOTCacheOutput=app.aot …` on a training run, then `java -XX:AOTCache=app.aot …` in production.
- JDK 25 (JEP 515): the cache also stores method profiles, so the JIT optimises sooner.

GraalVM Native Image instead compiles the whole app to a native executable before it runs: very fast start-up and small memory, at the cost of long builds and restrictions on reflection (inspecting classes and calling methods by name at run time) and dynamic class loading.

Trade-offs: interpreted vs JIT vs AOT

Approach	Start-up	Peak speed	Portability	Example
Pure interpretation	Instant	Slow	High	Early JVMs, CPython
JIT (HotSpot)	Slower (warm-up)	Very high — uses live profiles	High (bytecode)	Normal `java`
AOT cache (JDK 24+)	Faster	Same as JIT	High	`-XX:AOTCache`
Native image	Very fast	Good, often a bit lower without profile data	One binary per OS/CPU	GraalVM `native-image`

In the interview

Q: Is Java compiled or interpreted? Both. `javac` compiles source to bytecode ahead of time; the JVM interprets it and JIT-compiles hot methods to native code using profiling data — so long-running services get faster after warm-up.

- *Tiered compilation?* Interpreter → C1 (fast compile, profiling) → C2 (optimising compile using the profile).
- *Deoptimization?* Discarding compiled code whose speculative assumption became false and returning to the interpreter.
- *Why is a simple timing-loop microbenchmark misleading?* It mixes interpreted, C1 and C2 phases and the JIT may remove "useless" work; use JMH (Java Microbenchmark Harness; Chapter 16).

1.8 Memory and garbage collection — an overview

The picture you need now; algorithms, tuning and profiling are in Chapter 16.

What it is

> **Key term — thread:** one independent sequence of instructions inside a program. A program can run many at once (e.g. one per web request); each has its own call stack but shares the heap. Covered in the concurrency chapters.

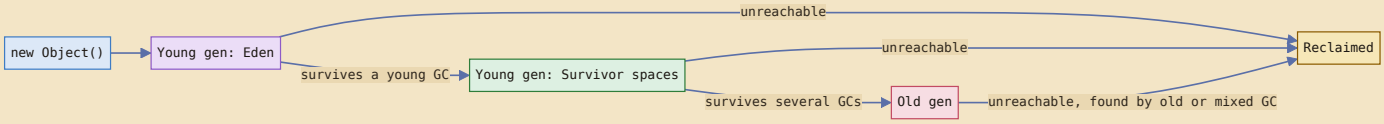
Area	Holds	Lifetime
Stack (one per thread)	*Frames* per method call: locals, parameters, partial results. Primitive locals live here; for objects only the **reference** (pointer-like handle).	Frame disappears when the method returns
Heap (shared)	All objects and arrays (`new …`).	Until the GC reclaims them
Metaspace	Class metadata (methods, field layouts), in native memory (outside the Java heap).	While the class loader is alive
Code cache	JIT-compiled machine code.	Until evicted/deoptimized

So in a method, `int x = 5;` lives on the stack; `new int[5]` lives on the heap with its reference on the stack.

Garbage collection (GC) automatically reclaims heap objects that are no longer **reachable** — no chain of references leads to them from a **GC root** (a local in an active frame, a static field, an active thread, a few JVM-internal references). Unreachable objects are garbage even if they point to each other, so reference cycles are collected.

How it works (the common core)

1. **Mark**: from the roots, follow references and mark every reachable object.
2. **Sweep / compact / copy**: free unmarked memory; many collectors also move live objects together to avoid fragmentation (free memory scattered in gaps too small to use).
3. **Generations**: most objects die young (the *weak generational hypothesis* — e.g. temporary strings in a request). Generational collectors split the heap into a **young generation**, collected often and cheaply (most of it is dead; only the few live objects are copied), and an **old generation** for survivors, collected rarely.
4. **Pauses**: some work stops application threads (**stop-the-world** pause); modern collectors do most work **concurrently** with your threads.



The collectors in JDK 25

Collector	Flag	Design	Good for
G1 (Garbage-First)	`-XX:+UseG1GC`	Generational, region-based (heap split into equal chunks), mostly concurrent marking; pause-time goal (default 200 ms)	**The default** — general servers and desktops
ZGC	`-XX:+UseZGC`	Concurrent, generational (the only mode since JDK 24); pauses typically under 1 ms regardless of heap size	Large heaps, latency-sensitive services (per-request time matters)
Shenandoah	`-XX:+UseShenandoahGC`	Concurrent compacting; generational mode became a product feature in JDK 25	Low latency; in many OpenJDK builds (not Oracle JDK)
Parallel	`-XX:+UseParallelGC`	Stop-the-world, many threads; maximises throughput (work done per second)	Batch jobs where pauses do not matter
Serial	`-XX:+UseSerialGC`	Single-threaded, stop-the-world	Tiny heaps, small containers
Epsilon	experimental	Allocates, never collects	Testing, very short-lived jobs

The JVM picks G1 unless the machine looks "small" (fewer than 2 CPUs or under 1792 MB), where it picks Serial; JDK 27 (JEP 523) makes G1 the default everywhere. Default max heap = ¼ of physical memory — `WhereAmI` below reports 12 288 MiB on a 48 GiB machine (1 MiB = 1024 × 1024 bytes, 1 GiB = 1024 MiB). **Latency** and **throughput** recur in System Design Fundamentals.

Watching the GC

`code/GarbageMaker.java` creates 200 000 short-lived 1 KiB arrays and keeps ten 1 MiB arrays:

```
List<byte[]> keep = new ArrayList<>();          // long-lived: stays reachable
for (int i = 0; i < 200_000; i++) {
    byte[] temp = new byte[1024];                // short-lived: garbage after this iteration
    temp[0] = (byte) i;
    created++;
    if (i % 20_000 == 0) keep.add(new byte[1024 * 1024]);
}
```

`java -Xlog:gc -Xmx256m -cp out ch01.gc.GarbageMaker` (selected lines from one run; timings and sequence vary):

```
[0.004s][info][gc] Using G1
[0.017s][info][gc] GC(0) Pause Young (Concurrent Start) (G1 Humongous Allocation) 1M->1M(10M) 0.407ms
[0.017s][info][gc] GC(1) Concurrent Mark Cycle
[0.018s][info][gc] GC(2) Pause Young (Normal) (G1 Evacuation Pause) 6M->3M(10M) 0.195ms
[0.022s][info][gc] GC(4) Pause Young (Prepare Mixed) (G1 Evacuation Pause) 22M->5M(30M) 0.580ms
[0.024s][info][gc] GC(5) Pause Young (Mixed) (G1 Evacuation Pause) 22M->5M(30M) 0.305ms
...
temporary arrays created: 200000
arrays still reachable  : 10
```

`22M->5M(30M) 0.580ms` = heap used before → after (current heap size), pause length. Young collections free most memory because the temporary arrays are dead. "Humongous" = an object at least half a G1 region (the 1 MiB arrays), allocated specially. With `-XX:+UseZGC` the first line reads `Using The Z Garbage Collector`.

Ask the JVM about itself with `code/WhereAmI.java`:

```
Runtime.Version v = Runtime.version();
System.out.println("Feature release : " + v.feature());
System.out.println("class file ver. : " + System.getProperty("java.class.version"));
System.out.println("Max heap (MiB)  : " + Runtime.getRuntime().maxMemory() / (1024 * 1024));
```

Output (this machine; yours will differ):

```
Feature release : 26
java.vendor     : Oracle Corporation
java.vm.name    : OpenJDK 64-Bit Server VM
class file ver. : 70.0
os.name         : Mac OS X
CPU cores seen  : 14
Max heap (MiB)  : 12288
```

Compiled for Java 25, running on JDK 26 — newer JVMs run older class files.

Common mistakes

- Believing Java cannot leak: a static `Map` cache that only grows keeps every entry reachable forever — a **memory leak**.
- Calling `System.gc()` to "free memory": only a request, usually harmful.
- Relying on `finalize()`: deprecated for removal (JEP 421, Java 18). Use try-with-resources (Chapter 8).
- `-Xmx` (max heap) larger than the container's memory limit (the JVM reads container limits automatically; let it, or size explicitly).

In the interview

Q: How does Java GC decide what to free? By reachability, not reference counting (counting references to each object): anything not reachable from GC roots (thread stacks, static fields, JNI handles — references held by native code via the Java Native Interface) is garbage, so cycles are collected. Most collectors are generational because most objects die young.

- *Default GC?* G1 since Java 9 on server-class machines.
- *When ZGC?* Large heaps or strict latency targets, where sub-millisecond pauses matter more than a little throughput.
- *Stack vs heap?* Stack: per-thread frames with locals, freed on return. Heap: shared objects, freed by GC.

TL;DR

Stack = method frames; heap = objects. GC frees unreachable objects; generational collectors exploit "most objects die young". Default G1; ZGC/Shenandoah for low latency; Parallel for throughput.

1.9 jshell — the Java REPL

What it is

`jshell` (Java 9, JEP 222) is a **REPL** — *Read-Evaluate-Print Loop*: type a snippet, see the result, no class or `main` needed. A scratchpad for trying APIs; anything you want to keep, test or share belongs in a source file.

A session

```
|  Welcome to JShell -- Version 26
|  For an introduction type: /help intro

jshell> int x = 21
x ==> 21

jshell> x * 2
$2 ==> 42

jshell> "Java".length()
$3 ==> 4

jshell> String shout(String s) { return s.toUpperCase() + "!"; }
|  created method shout(String)

jshell> shout("hi")
$5 ==> "HI!"

jshell> /vars
|    int x = 21
|    int $2 = 42
|    int $3 = 4
|    String $5 = "HI!"
```

- Unassigned expressions get reusable scratch names (`$2`, `$3`…).
- Semicolons are optional for single statements; methods can be declared at top level and redefined.

Command	Does
`/help`, `/help intro`	Help
`/vars`, `/methods`, `/types`, `/imports`	List definitions/imports
`/list`	Show your snippets
`/edit <name>`	Open a snippet in an editor
`/save file.jsh`, `/open file.jsh`	Save/load a session
`/reset`, `/exit`	Clear state, quit
Tab	Completion and method signatures

On JDK 26, `/imports` shows `import java.base`, so every public class in `java.base` is usable without imports.

1.10 Launching source files directly

Single-file and multi-file source launch

Since **Java 11** (JEP 330) `java Hello.java` compiles a single file in memory and runs it; no `.class` files are written. Since **Java 22** (JEP 458) the program may use other source files in the same directory tree, compiled on demand.

`code/launch/Main.java` uses `code/launch/Greeter.java`:

```
// Main.java
public class Main {
    public static void main(String[] args) {
        System.out.println(Greeter.greet("launcher"));
    }
}
// Greeter.java
public class Greeter {
    static String greet(String who) {
        return "Hello from " + who + ", compiled in memory";
    }
}
```

```
cd code/launch && java Main.java
```

Output:

```
Hello from launcher, compiled in memory
```

On Linux/macOS a single-file program can start with a **shebang** line (`#!` first line naming the program to run the file), e.g. `#!/usr/bin/env -S java --source 25`, in a file *without* the `.java` extension, and run as a script.

Compact source files and instance `main` — Java 25

JEP 512 (final in **Java 25**, after previews — opt-in trial versions, 1.12 — in 21–24):

- A **compact source file** may declare methods and fields with no enclosing class; the compiler wraps them in an implicit `final` class (cannot be extended).
- `main` may be a parameterless **instance method** (runs on an object, not the class): `void main()`.
- It automatically imports every public class of `java.base` (as if it began with `import module java.base;`), so `List`, `Map`, `Path` need no imports.
- New class **`java.lang.IO`**: `IO.println(…)`, `IO.print(…)`, `IO.println()`, `IO.readln(prompt)`/`IO.readln()` for simple console I/O (built on `System.out` and `System.in`).

`code/compact/HelloCompact.java` — the whole file:

```
void main() {
    List<String> langs = List.of("Java", "Kotlin", "Scala");   // java.util.List without an import
    IO.println("Hello from a compact source file!");
    IO.println("JVM languages: " + langs);
}
```

```
cd code/compact && java HelloCompact.java
```

Output:

```
Hello from a compact source file!
JVM languages: [Java, Kotlin, Scala]
```

How the launcher chooses `main`: `main(String[])` if present, else a no-argument `main()`. If the chosen method is not `static`, it first creates an instance with the no-argument constructor.

Trade-offs

Style	Best for	Limits
`javac` + `java`	Real applications, packages, dependencies	Two steps
`java File.java`	Scripts, examples, teaching, small tools	Compiles every run; no dependency management
Compact source file	First programs, quick utilities	No `package`; cannot be referenced by name from other code; becomes a normal class as it grows

Common mistakes

- Adding a `package` declaration to a compact source file (not allowed).
- Expecting instance `main` on Java 21 without `--enable-preview` — preview until Java 25.

1.11 IDEs, editors and build tools

What they are

An **IDE** (Integrated Development Environment) combines editor, compiler integration, debugger (pauses a running program to step through and inspect variables), refactoring and test runner: **IntelliJ IDEA** (JetBrains; most popular, strongest refactoring; free and paid tiers), **Eclipse** (free, open source, its own incremental compiler), **VS Code** + *Extension Pack for Java* (lightweight, mixed-language work), **Apache NetBeans** (free, open source).

Build tools automate compiling, downloading **dependencies** (external libraries your project uses, at the right versions), testing and packaging: **Maven** (XML `pom.xml`, fixed lifecycle) and **Gradle** (Groovy/Kotlin DSL — a small build-script language; flexible and incremental, rebuilding only what changed). Both read `JAVA_HOME` and support *toolchains* to pin a JDK version. Details: Chapter 18.

Suggested path: plain `javac`/`java`/`jshell` for Chapters 1–3; an IDE from Chapter 5 (multiple classes, packages); Maven/Gradle for anything with dependencies.

Coding conventions (used throughout this book)

Kind	Convention	Example
Class, interface, record, enum	`UpperCamelCase` noun	`OrderService`, `Shape`
Method, variable	`lowerCamelCase`; methods are verbs	`computeTotal()`, `itemCount`
Constant (`static final`)	`UPPER_SNAKE_CASE`	`MAX_RETRIES`
Package	all lowercase, reversed domain	`com.example.billing`
Indentation	4 spaces; opening brace on the same line	`if (x) {`

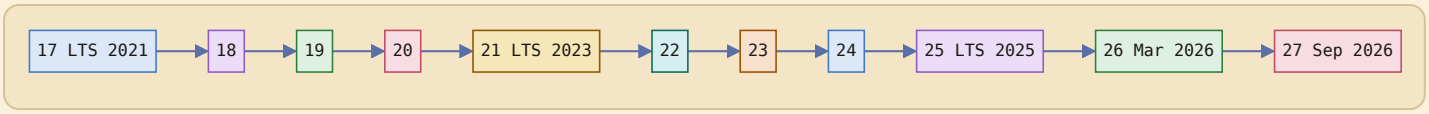
1.12 Java releases, LTS and JEPs

How the release model works

- A **feature release** ships every six months, **March** and **September** (since Java 10, 2018). Version numbers are counters: 25, 26, 27…
- Each gets a couple of quarterly **update releases** (25.0.1, 25.0.2) from Oracle's OpenJDK builds; vendors keep patching the **LTS** ones.
- **Long-Term Support (LTS)** releases: **8, 11, 17, 21, 25** — every two years since 17; next expected September 2027 (Java 29). Oracle offers at least eight years of LTS support (five Premier + three

Extended).

- Non-LTS releases (22, 23, 24, 26, 27…) are production-quality but supported only until the next release.



JEPs and feature maturity

New features are proposed in **JEPs** (JDK Enhancement Proposals, openjdk.org/jeps), e.g. *JEP 512: Compact Source Files and Instance Main Methods*. Stages:

Stage	Meaning	How to use it
Incubator	Experimental API in a `jdk.incubator.*` module (e.g. the Vector API, still incubating in 27)	`--add-modules jdk.incubator.vector`
Preview	Complete language/VM/API feature that may still change	`javac --release 25 --enable-preview …` and `java --enable-preview …`
Final / permanent	Standard, stable part of Java	Just use it
Withdrawn	Removed before final (e.g. *string templates*, previewed 21–22, dropped in 23)	—

This book marks each modern feature with the version that **finalised** it (e.g. *records — Java 16*). Full tour: Chapter 17.

Which version should I use?

Situation	Pick
New project, or learning	**Java 25 LTS**
Existing Java 17/21 service	Stay while supported; plan an upgrade to 25 (mostly smooth — run tests on the new JDK first)
Legacy Java 8	8 → 11 → 17 → 21/25 needs attention to removed Java EE modules and strong encapsulation of JDK internals (code can no longer reach the JDK's private implementation classes)
Library you publish	Compile with `--release` of the *oldest* Java you support

In the interview

Q: What is an LTS release and why does it matter? A feature release vendors support for many years (8, 11, 17, 21, 25); companies standardise on them to avoid upgrading every six months.

- *How often does Java release?* Every six months (March/September); LTS every two years. Java 26 is not LTS.
- *Preview feature?* Finished-but-not-final; opt in with `--enable-preview` on both `javac` and `java`; may change or be withdrawn.
- *Name three Java 21 features.* Virtual threads, pattern matching for `switch`, record patterns, sequenced collections.

Chapter TL;DR

- **Java** = statically typed, object-oriented language + portable runtime. Current LTS **Java 25** (Sept 2025); releases every March/September, LTS every two years.
- **JDK ⊃ runtime (JRE) ⊃ JVM.** Install a JDK; ship a `jlink` runtime or container image.

- `javac` compiles `.java` → `.class` bytecode (major version 69 = Java 25); `java` runs it. `--release N` targets an older Java safely.
- The JVM **loads lazily, verifies, interprets, then JIT-compiles** hot code (C1 → C2) using profiles; AOT caches (JDK 24–25) and native images cut warm-up.
- **Stack** = frames and locals; **heap** = objects; **GC** frees unreachable objects. Default G1; ZGC/Shenandoah for low latency.
- **jshell** for experiments; `java File.java` runs sources directly (multi-file since 22); **compact source files + `void main()` + `IO`** final in **Java 25**.
- Exam traps: OCP Chapter 1. Next: Chapter 2 — Basics: types, variables and operators.